UC Berkeley

Department of Electrical Engineering and Computer Sciences

EECS 127 - EECS 227AT. Optimization Models in Engineering

Lecture Note 18

Fall 2025

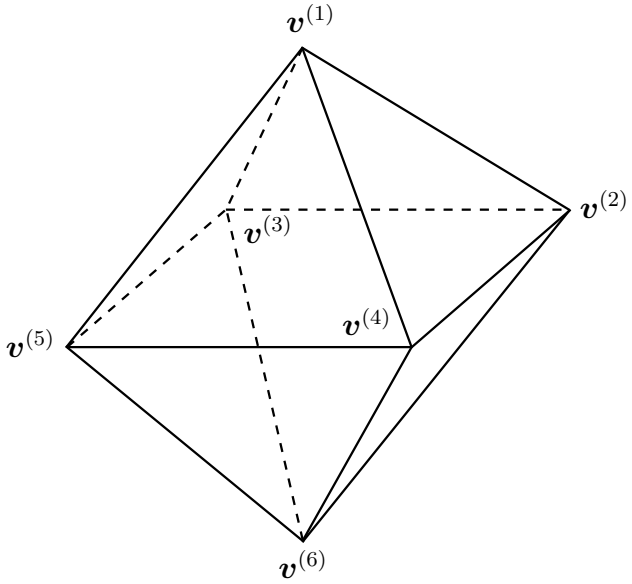
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If a standard LP is non-degenerate, each vertex has $n - m$ zeros and m non-zero elements.

Example 1 (Adjacent vertices)**.** Consider a three dimensional polyhedron:



- Geometrically, neighbors of $\boldsymbol{v}^{(1)}$ are $\boldsymbol{v}^{(2)}, \boldsymbol{v}^{(3)}, \boldsymbol{v}^{(4)}, \boldsymbol{v}^{(5)}$ but not $\boldsymbol{v}^{(6)}$.
- You can move from a vertex to its neighbor via a straight line on the boundary.

■

Theorem 1. *Consider a standard non-degenerate LP. Let $\boldsymbol{v}^{(1)}, \ldots, \boldsymbol{v}^{(P)}$ be vertices of its feasible set. Two vertices $\boldsymbol{v}^{(i)}$ and $\boldsymbol{v}^{(j)}$ are adjacent iff there are $n - m - 1$ indices in the set $\{1, \ldots, n\}$ at which $\boldsymbol{v}^{(i)}$ and $\boldsymbol{v}^{(j)}$ are both zero.*

Example 2. Let $n = 6$, $m = 2$, and consider

$$\boldsymbol{v}^{(1)} = \begin{bmatrix} 1 \\ 0 \\ 2 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \quad \boldsymbol{v}^{(2)} = \begin{bmatrix} 5 \\ 0 \\ 0 \\ 3 \\ 0 \\ 0 \end{bmatrix}.$$

We have $n - m - 1 = 6 - 2 - 1 = 3$, and $\boldsymbol{v}^{(1)}, \boldsymbol{v}^{(2)}$ are both zero at indices $2, 5, 6$. Therefore, by Theorem 1, $\boldsymbol{v}^{(1)}$ and $\boldsymbol{v}^{(2)}$ are adjacent. ■

Remark: The polyhedron in Example 1 is not the feasible set of any *standard* LP, so Theorem 1 doesn't apply to it. In a true 3-variable LP in standard form (i.e., $\min \boldsymbol{a}_0^{\mathrm{T}} \boldsymbol{x}$ subject to $\mathbf{A}\boldsymbol{x} = \boldsymbol{b}$ and $\boldsymbol{x} \geq \mathbf{0}$), the feasible region is the intersection of the affine subspace defined by the equality constraints with the non-negative orthant. Each equality constraint $\boldsymbol{a}_i^{\mathrm{T}} \boldsymbol{x} = b_i$ restricts the solution to a (at most) 2-dimensional plane in $\mathbb{R}^3$, and the additional non-negativity constraints truncate that plane to form a convex polyhedron (or possibly lower-dimensional face).

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How to solve LP. Simplex algorithm:

(1) Find an arbitrary vertex and set the parameter $\boldsymbol{y}$ to be that vertex.

(2) Check all neighbors of vertex $\boldsymbol{y}$ to see if there is an adjacent vertex $\tilde{\boldsymbol{y}}$ such that $\boldsymbol{a}_0^{\mathrm{T}}\tilde{\boldsymbol{y}} < \boldsymbol{a}_0^{\mathrm{T}}\boldsymbol{y}$.

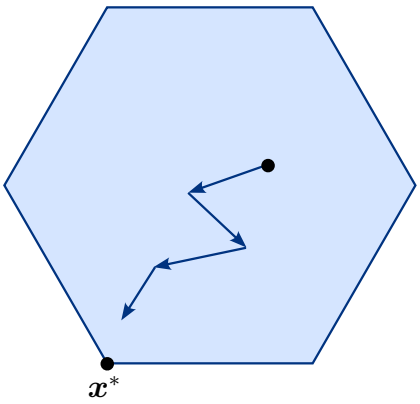
(3) If such adjacent neighbor $\tilde{\boldsymbol{y}}$ doesn't exist, $\boldsymbol{y}$ is an optimal solution to LP and we stop the algorithm.

(4) If such adjacent neighbor $\tilde{\boldsymbol{y}}$ exists, update parameter $\boldsymbol{y}$ to be $\tilde{\boldsymbol{y}}$ and jump to step (2).

The details of simplex algorithm and the degenerate case are out of scope for this class, but you should learn the high-level idea.

Example 3. Consider the feasible set shown in Example 1. Assume $\boldsymbol{v}^{(6)}$ is optimal and $\boldsymbol{a}_0^{\mathrm{T}}\boldsymbol{v}^{(2)} < \boldsymbol{a}_0^{\mathrm{T}}\boldsymbol{v}^{(1)}$. By running the simplex method, we could start at $\boldsymbol{v}^{(1)}$, then move to $\boldsymbol{v}^{(2)}$, then move to $\boldsymbol{v}^{(6)}$ and stop. ■

- The simplex method has been the most fundamental algorithm for LP. It works well in many different applications, but theoretically it could be very slow since the number of vertices is exponential and we may need to visit an exponential number of vertices to get to an optimal vertex.
- The interior point method has addressed this issue but it is beyond the scope of this class.
- High-level idea:



  Start in the interior of the polyhedron and move towards the optimal solution, rather than going on the boundary and checking many vertices.

- Most real-world problems are not LP, but they can be approximated by or reformulated as LP.

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Epigraph is a powerful technique for LP conversion.

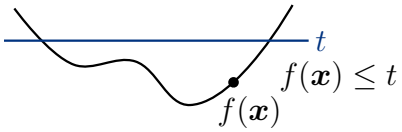
- Consider a general optimization:

$$\begin{aligned} \min_{\boldsymbol{x}\in\mathbb{R}^n} \quad & f(\boldsymbol{x}), \\ \text{s.t.} \quad & \boldsymbol{x} \in \mathcal{S}. \end{aligned}$$

  This is equivalent to

$$\begin{aligned} \min_{\boldsymbol{x}\in\mathbb{R}^n, t\in\mathbb{R}} \quad & t, \\ \text{s.t.} \quad & \boldsymbol{x} \in \mathcal{S}, \\ & f(\boldsymbol{x}) \le t. \end{aligned}$$

- Here we have included a new variable t (called *slack variable*) that serves as an upper bound on the function at $\boldsymbol{x}$, and we minimize over the upper bound.



- By minimizing t, we push the bound down until $f(\boldsymbol{x}) = t$ and that means we are minimizing $f(\boldsymbol{x})$.
- This is the *epigraph technique*.

Example 4 (Epigraph technique)**.** Consider

$$\begin{aligned} \min_{\boldsymbol{x}\in\mathbb{R}^n} \quad & \|\boldsymbol{x}\|_\infty, \\ \text{s.t.} \quad & \mathbf{A}\boldsymbol{x} = \boldsymbol{b}, \\ & \mathbf{C}\boldsymbol{x} \le \boldsymbol{d}. \end{aligned}$$

This is not LP, but if we use epigraph:

$$\begin{aligned} \min_{\boldsymbol{x}\in\mathbb{R}^n, t\in\mathbb{R}} \quad & t, \\ \text{s.t.} \quad & \mathbf{A}\boldsymbol{x} = \boldsymbol{b}, \\ & \mathbf{C}\boldsymbol{x} \le \boldsymbol{d}, \\ & \|\boldsymbol{x}\|_\infty \le t. \end{aligned}$$

Note that

$$\begin{aligned} \|\boldsymbol{x}\|_\infty \le t \;&\Leftrightarrow\; \max(|x_1|, \ldots, |x_n|) \le t \\ &\Leftrightarrow\; |x_i| \le t,\ i = 1, \ldots, n \\ &\Leftrightarrow\; -t \le x_i \le t,\ i = 1, \ldots, n. \end{aligned}$$

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Therefore, the above optimization is equivalent to LP:

$$\begin{aligned} \min_{\boldsymbol{x}\in\mathbb{R}^n, t\in\mathbb{R}} \quad & t, \\ \text{s.t.} \quad & \mathbf{A}\boldsymbol{x} = \boldsymbol{b}, \\ & \mathbf{C}\boldsymbol{x} \le \boldsymbol{d}, \\ & -t \le x_i \le t,\ i = 1, \dots, n. \end{aligned}$$

■

Example 5. Consider

$$\begin{aligned} \min_{\boldsymbol{x}\in\mathbb{R}^n} \quad & \|\boldsymbol{x}\|_1, \\ \text{s.t.} \quad & \mathbf{A}\boldsymbol{x} = \boldsymbol{b}, \\ & \mathbf{C}\boldsymbol{x} \le \boldsymbol{d}. \end{aligned}$$

The previous epigraph does not work directly since $\|\boldsymbol{x}\|_1 \le t \Leftrightarrow |x_1| + \cdots + |x_n| \le t$, and this cannot be converted to linear constraints.
We write the objective as $|x_1| + \cdots + |x_n|$ and define an upper bound for each term:

$$\begin{aligned} \min_{\boldsymbol{x},\boldsymbol{t}\in\mathbb{R}^n} \quad & t_1 + \cdots + t_n, \\ \text{s.t.} \quad & \mathbf{A}\boldsymbol{x} = \boldsymbol{b}, \\ & \mathbf{C}\boldsymbol{x} \le \boldsymbol{d}, \\ & |x_i| \le t_i,\ i = 1, \dots, n. \end{aligned}$$

The new problem is LP (note that $|x_i| \le t_i \Leftrightarrow -t_i \le x_i \le t_i$).
Similar objectives like $\|\boldsymbol{x}\|_1 + \|\boldsymbol{x}\|_\infty$ can be converted to LP as well. ■

Another special class of convex optimization is *quadratic program* (QP).

- Change the linear objective of LP to a convex and quadratic function to get QP:

$$\begin{aligned} \min_{\boldsymbol{x}\in\mathbb{R}^n} \quad & \boldsymbol{x}^{\mathrm{T}}\mathbf{P}_0\boldsymbol{x} + \boldsymbol{q}_0^{\mathrm{T}}\boldsymbol{x} + r_0, \\ \text{s.t.} \quad & \boldsymbol{a}_i^{\mathrm{T}}\boldsymbol{x} = b_i,\ i = 1, \dots, m, \\ & \boldsymbol{c}_j^{\mathrm{T}}\boldsymbol{x} \le d_j,\ j = 1, \dots, k, \end{aligned}$$

  where $r_0 \in \mathbb{R}$, $\boldsymbol{q}_0 \in \mathbb{R}^n$, $\mathbf{P}_0 \in \mathbb{S}^n$ (the set of symmetric matrices).

- The Hessian of the objective is $2\mathbf{P}_0$. Convexity requires $\mathbf{P}_0 \succeq \mathbf{0}$.

- If $\mathbf{P}_0 = \mathbf{0}$, QP reduces to LP. Thus,

$$\text{LP} \subset \text{QP} \subset \text{convex optimization}.$$

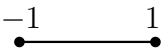
- The feasible set of QP is a polyhedron, like LP.

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- However, QP is harder to solve than LP.
- For example, if an LP with $n = 10,000$ can be solved in 5 minutes on a laptop, the QP can be solved with only $n = 1,000$ in 5 minutes.
- To explain the reason, consider

$$\begin{aligned} \min_{x\in\mathbb{R}} \quad & \alpha x^2 + \beta x, \\ \text{s.t.} \quad & -1 \le x \le 1, \end{aligned}$$

where α, β are constant. The feasible set has two vertices:



(1) Assume $\alpha = 0$, then the solution is either vertex $x = 1$ or vertex $x = -1$.

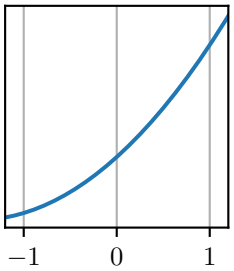
If $\beta > 0$, $x^* = -1$;
If $\beta < 0$, $x^* = 1$;
If $\beta = 0$, $\mathcal{X}^{\text{opt}} = [-1, 1]$.

(2) Assume $\alpha = 1$, then

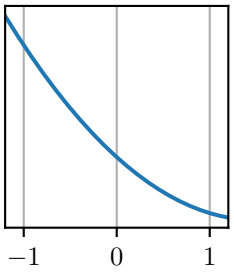
$$\frac{\partial(x^2 + \beta x)}{\partial x} = 2x + \beta.$$

The derivative is zero at $x = -\beta/2$.

If $\beta > 2$, $x^* = -1$:

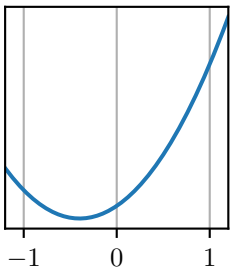


If $\beta < -2$, $x^* = 1$:



If $\beta \in [-2, 2]$, $x^* = -\beta/2$:

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As β goes from -2 to 2, the solution goes from vertex 1 to vertex -1.

- In general, solution of QP could be any point in the polyhedron depending on the objective, so searching over a finite number of vertices is not enough.

Another special class of convex optimization is *quadratically constrained quadratic program* (QCQP).

- To obtain QCQP, allow inequalities of QP to be convex and quadratic:

$$\begin{aligned} \min_{\boldsymbol{x}\in\mathbb{R}^n} \quad & \boldsymbol{x}^{\mathrm{T}}\mathbf{P}_0\boldsymbol{x} + \boldsymbol{q}_0^{\mathrm{T}}\boldsymbol{x} + r_0, \\ \text{s.t.} \quad & \boldsymbol{x}^{\mathrm{T}}\mathbf{P}_j\boldsymbol{x} + \boldsymbol{q}_j^{\mathrm{T}}\boldsymbol{x} + r_j \le 0,\ j = 1,\dots,k, \\ & \boldsymbol{a}_i^{\mathrm{T}}\boldsymbol{x} = b_i,\ i = 1,\dots,m, \end{aligned}$$

  where $\mathbf{P}_0, \mathbf{P}_1, \dots, \mathbf{P}_k$ are PSD matrices (to guarantee convexity), $\boldsymbol{q}_0, \dots, \boldsymbol{q}_k$ are vectors, and $r_0, \dots, r_k$ are scalars.

- If $\mathbf{P}_1, \dots, \mathbf{P}_k$ are set to zero, QCQP reduces to QP. Thus,

$$\text{LP} \subset \text{QP} \subset \text{QCQP} \subset \text{convex optimization}.$$

- The feasible set of QCQP is

$$\left\{ \boldsymbol{x} \in \mathbb{R}^n \;\middle|\; \underbrace{\boxed{\boldsymbol{x}^{\mathrm{T}}\mathbf{P}_j\boldsymbol{x} + \boldsymbol{q}_j^{\mathrm{T}}\boldsymbol{x} + r_j \le 0}}_{\text{ellipsoid}},\ j = 1,\dots,k;\ \underbrace{\boxed{\boldsymbol{a}_i^{\mathrm{T}}\boldsymbol{x} = b_i}}_{\text{hyperplane}},\ i = 1,\dots,m \right\}.$$

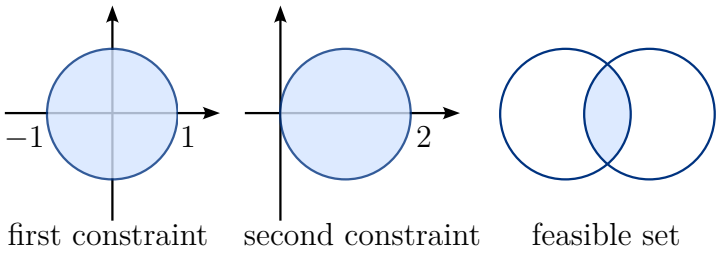
  It is an intersection of ellipsoids and hyperplanes.

- For example, consider the following feasible set:

$$\left\{ \boldsymbol{x} \in \mathbb{R}^2 \;\middle|\; \begin{array}{l} [x_1, x_2] \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + [0, 0] \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} - 1 \le 0 \\ [x_1, x_2] \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + [-2, 0] \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + 0 \le 0 \end{array} \right\}.$$

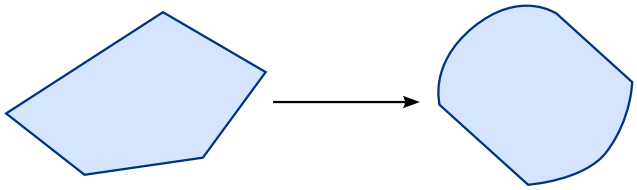
  It is the intersection of two balls, $x_1^2 + x_2^2 \le 1$ and $(x_1 - 1)^2 + x_2^2 \le 1$:

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All points on the boundary are extreme points. There are two sharp points at the top and bottom.

- The feasible set of QCQP is like a polyhedron, but its boundary consists of curly faces rather than flat faces:



Due to the shape of boundary, QCQP is slower than QP to solve numerically.

Applications: Many problems in the world are non-convex but we approximate them with LP/QP/QCQP.

- Consider the optimization

$$\begin{aligned} \min \quad & f(\boldsymbol{x}), \\ \text{s.t.} \quad & \boldsymbol{x} \in \mathcal{S}. \end{aligned}$$

- Assume f is a convex function and $\mathcal{S}$ is a non-convex set.
- Consider a convex set $\tilde{\mathcal{S}}$ such that $\mathcal{S} \subseteq \tilde{\mathcal{S}}$.
- The problem

$$\begin{aligned} \min \quad & f(\boldsymbol{x}), \\ \text{s.t.} \quad & \boldsymbol{x} \in \tilde{\mathcal{S}} \end{aligned}$$

is called a *convex relaxation* of

$$\begin{aligned} \min \quad & f(\boldsymbol{x}), \\ \text{s.t.} \quad & \boldsymbol{x} \in \mathcal{S}. \end{aligned}$$

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Theorem 2 (Convex relaxation)**.** *Consider the optimization problem*

$$\begin{aligned} \min \quad & f(\boldsymbol{x}), \\ s.t. \quad & \boldsymbol{x} \in \mathcal{S}, \end{aligned} \tag{1}$$

where f is convex and $\mathcal{S}$ is non-convex. Let its convex relaxation be

$$\begin{aligned} \min \quad & f(\boldsymbol{x}), \\ s.t. \quad & \boldsymbol{x} \in \tilde{\mathcal{S}}. \end{aligned} \tag{2}$$

Let $\boldsymbol{x}^*$ and $\tilde{\boldsymbol{x}}$ be arbitrary global minima of (1) and (2). We have

(i) $f(\tilde{\boldsymbol{x}}) \leq f(\boldsymbol{x}^*)$, meaning that the convex relaxation provides a lower bound on the original problem.

(ii) If $\tilde{\boldsymbol{x}} \in \mathcal{S}$, then $\tilde{\boldsymbol{x}}$ is a global minimum for the original problem (1). (Note: $\tilde{\boldsymbol{x}}$ doesn't need to be equal to $\boldsymbol{x}^*$ since the solution may not be unique.)

Proof.

(i) Since $\tilde{\boldsymbol{x}}$ is a global minimum for (2), we have

$$f(\tilde{\boldsymbol{x}}) \leq f(\boldsymbol{x}), \quad \forall \boldsymbol{x} \in \tilde{\mathcal{S}}. \tag{3}$$

Also, $\mathcal{S} \subset \tilde{\mathcal{S}}$ and $\boldsymbol{x}^* \in \mathcal{S}$ since $\boldsymbol{x}^*$ is a solution for (1). Plug $\boldsymbol{x} = \boldsymbol{x}^*$ into (3) proves part (i).

(ii) For part (ii), since $\mathcal{S} \subset \tilde{\mathcal{S}}$, we obtain from (3) that

$$f(\tilde{\boldsymbol{x}}) \leq f(\boldsymbol{x}), \quad \forall \boldsymbol{x} \in \mathcal{S}.$$

Since $\tilde{\boldsymbol{x}} \in \mathcal{S}$ by assumption, the above inequality implies that $\tilde{\boldsymbol{x}}$ is a global minimum for (1).

■

Consider two optimization problems:

$$\begin{aligned} \min \quad & f_0(\boldsymbol{x}), \\ \text{s.t.} \quad & f_i(\boldsymbol{x}) \leq 0, \ i = 1, \ldots, m, \\ & h_j(\boldsymbol{x}) = 0, \ j = 1, \ldots, k \end{aligned} \tag{4}$$

and

$$\begin{aligned} \min \quad & f_0(\boldsymbol{x}), \\ \text{s.t.} \quad & \tilde{f}_i(\boldsymbol{x}) \leq 0, \ i = 1, \ldots, \tilde{m}, \\ & \tilde{h}_j(\boldsymbol{x}) = 0, \ j = 1, \ldots, \tilde{k}. \end{aligned} \tag{5}$$

- Note that $\tilde{m}, \tilde{k}$ could be different from m, k.

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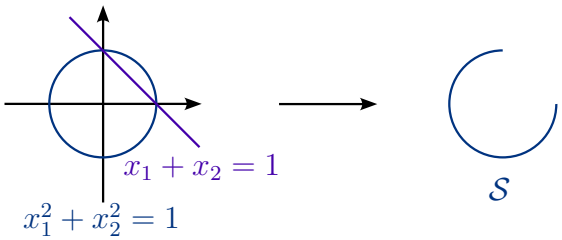
- If (5) is a convex optimization and if the feasible set of (4) is a subset of the feasible set of (5), then (5) is called a *convex relaxation* of (4), and Theorem 2 about lower bounding the solution holds.

Note: (4) is an algebraic representation of (1) and (5) is an algebraic representation of (2).

Example 6 (Convex relaxation)**.** Consider

$$\begin{aligned} \min \quad & \mathrm{e}^{x_1-x_2}+\mathrm{e}^{-x_1+x_2}, \\ \text{s.t.} \quad & x_1^2+x_2^2=1, \\ & x_1+x_2\le 1. \end{aligned} \tag{6}$$

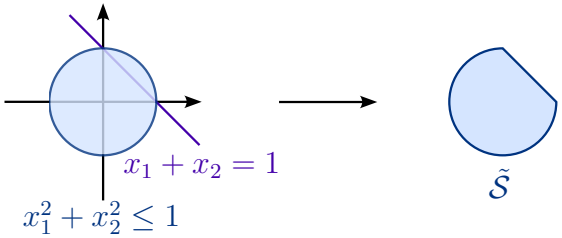
Its feasible set is a curve:



Now consider

$$\begin{aligned} \min \quad & \mathrm{e}^{x_1-x_2}+\mathrm{e}^{-x_1+x_2}, \\ \text{s.t.} \quad & x_1^2+x_2^2\le 1, \\ & x_1+x_2\le 1. \end{aligned} \tag{7}$$

Its feasible set is convex:



Since the new optimization (7) is convex and $\mathcal{S}\subset\tilde{\mathcal{S}}$, the new problem is a convex relaxation of (6).
The idea about changing equalities like $x_1^2+x_2^2=1$ to inequalities like $x_1^2+x_2^2\le 1$ is a popular method for getting convex relaxation. ■

Convex relaxation is very important in *integer programming* (IP).

- LP subject to the additional constraint that all elements of $\boldsymbol{x}$ are integers is called IP.
- IP in standard form:

$$\begin{aligned} \min_{\boldsymbol{x}\in\mathbb{R}^n} \quad & \boldsymbol{a}_0^{\mathrm{T}}\boldsymbol{x}, \\ \text{s.t.} \quad & \mathbf{A}\boldsymbol{x}=\boldsymbol{b}, \\ & \boldsymbol{x}\ge\mathbf{0}, \\ & x_i\in\mathbb{Z},\ i=1,\dots,n. \end{aligned}$$

- IP is non-convex.

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